Solutions to Chapter 2 Exercises

Chaos: An Introduction to Dynamical Systems (Alligood, Sauer, Yorke)

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Throughout, the Hénon map (2.27) is $f_{a,b}(x,y) = (a - x^2 + by,\ x)$ with $Df = \begin{pmatrix} -2x & b \\ 1 & 0 \end{pmatrix}$, and the cat map (2.43) is $S(v) = Av \pmod 1$ with $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$.

Exercise 2.1

Find the eigenvalues from $\lambda^2 - (\mathrm{tr})\lambda + \det = 0$.

(a) $\mathrm{tr} = 7$, $\det = -18$: $\lambda = 9, -2$. Both have magnitude > 1, so the origin is a **source**.

(b) $\mathrm{tr} = \frac{7}{4}$, $\det = \frac{5}{8}$: $\lambda = 1.25, 0.5$. One is > 1 and one < 1, so the origin is a **saddle**.

(c) $\mathrm{tr} = 1.2$, $\det = 0.32$: $\lambda = 0.8, 0.4$. Both have magnitude < 1, so the origin is a **sink**.

Exercise 2.2

Let $A = \begin{pmatrix} 4.5 & 8 \\ -2 & -3.5 \end{pmatrix}$. Then $\mathrm{tr}\, A = 1$ and $\det A = 0.25$, so $\lambda^2 - \lambda + 0.25 = (\lambda - 0.5)^2$, a repeated eigenvalue 0.5. Let $N = A - 0.5I = \begin{pmatrix} 4 & 8 \\ -2 & -4 \end{pmatrix}$. Then $N^2 = 0$, so

$$A^n = (0.5I + N)^n = 0.5^n I + n\, 0.5^{n-1} N \longrightarrow 0.$$

Hence the limit is $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

Exercise 2.3

$g(x,y) = (x^2 - 5x + y,\ x^2)$. Fixed points: $y = x^2$ and $x = x^2 - 5x + y = 2x^2 - 5x$, so $2x^2 - 6x = 0$, giving $x = 0$ or 3. The fixed points are $(0,0)$ and $(3,9)$. The Jacobian is $Dg = \begin{pmatrix} 2x - 5 & 1 \\ 2x & 0 \end{pmatrix}$.

- At $(0,0)$: $\begin{pmatrix} -5 & 1 \\ 0 & 0 \end{pmatrix}$ has eigenvalues $-5, 0$, so it is a **saddle**.
- At $(3,9)$: $\begin{pmatrix} 1 & 1 \\ 6 & 0 \end{pmatrix}$ has $\lambda^2 - \lambda - 6 = 0$, so $\lambda = 3, -2$. Both have magnitude > 1, so it is a **source**.

Exercise 2.4

Fixed points. We need $y = x$ and $x = a - x^2 + bx$, i.e. $x^2 + (1-b)x - a = 0$. The eigenvalues of Df satisfy $\lambda^2 + 2x\lambda - b = 0$.

Period-two orbits. A period-two orbit is $\{(x_1, x_2), (x_2, x_1)\}$ with $x_1 \neq x_2$, where $x_2 = a - x_1^2 + bx_2$ and $x_1 = a - x_2^2 + bx_1$. Subtracting gives $x_1 + x_2 = 1 - b$; adding then gives $x_1 x_2 = (1-b)^2 - a$. So x_1, x_2 are the roots of

$$t^2 - (1-b)t + (1-b)^2 - a = 0,$$

which are real and distinct iff $a > \frac{3}{4}(1-b)^2$. For such an orbit, the product of Jacobians is $\begin{pmatrix} 4x_1x_2 + b & -2bx_2 \\ -2x_1 & b \end{pmatrix}$, with trace $T = 4x_1x_2 + 2b$ and determinant b^2.

(a) $a = -0.56$, $b = -0.5$. Fixed points: $x^2 + 1.5x + 0.56 = 0$, so $x = -0.7, -0.8$.

- $(-0.7, -0.7)$: $\lambda^2 - 1.4\lambda + 0.5 = 0$ gives $\lambda = 0.7 \pm 0.1i$, $|\lambda| = \sqrt{0.5} < 1$: **sink**.
- $(-0.8, -0.8)$: $\lambda^2 - 1.6\lambda + 0.5 = 0$ gives $\lambda \approx 1.174, 0.426$: **saddle**.

Period two: $\frac{3}{4}(1.5)^2 = 1.6875 > a$, so there are **no** period-two orbits.

(b) $a = 0.21$, $b = 0.6$. Fixed points: $x^2 + 0.4x - 0.21 = 0$, so $x = 0.3, -0.7$.

- $(0.3, 0.3)$: $\lambda^2 + 0.6\lambda - 0.6 = 0$ gives $\lambda \approx 0.531, -1.131$: **saddle**.
- $(-0.7, -0.7)$: $\lambda^2 - 1.4\lambda - 0.6 = 0$ gives $\lambda \approx 1.744, -0.344$: **saddle**.

Period two: $\frac{3}{4}(0.4)^2 = 0.12 < a$, so an orbit exists. Here $t^2 - 0.4t - 0.05 = 0$ gives $t = 0.5, -0.1$, so the orbit is $\{(0.5, -0.1), (-0.1, 0.5)\}$. Then $T = 4(-0.05) + 1.2 = 1$ and $\det = 0.36$, so $\lambda^2 - \lambda + 0.36 = 0$ gives $\lambda = 0.5 \pm \frac{\sqrt{11}}{10}i$, with $|\lambda| = 0.6 < 1$: the orbit is a **sink**.

Exercise 2.5

$f(x, y, z) = (x^2 y,\ y^2,\ xz + y)$. From $y = y^2$ we get $y = 0$ or 1.

- $y = 0$: $x = 0$, and then $z = 0 \cdot z + 0 = 0$. This gives $(0, 0, 0)$.
- $y = 1$: $x = x^2$, so $x = 0$ or 1. If $x = 1$ then $z = z + 1$, impossible. If $x = 0$ then $z = 1$. This gives $(0, 1, 1)$.

The Jacobian is $Df = \begin{pmatrix} 2xy & x^2 & 0 \\ 0 & 2y & 0 \\ z & 1 & x \end{pmatrix}$.

- At $(0,0,0)$ all eigenvalues are 0: **sink**.
- At $(0,1,1)$ the matrix is lower triangular with eigenvalues 0, 2, 0: **saddle**.

Exercise 2.6

$f(x,y) = (\sin\frac{\pi}{3}x,\ y/2)$. Fixed points: $y = 0$ and $x = \sin(\pi x/3)$. Since $|\sin| \le 1$, $|x| \le 1$. The function $s(x) - x$ with $s(x) = \sin(\pi x/3)$ is strictly concave on $[0,3]$ and vanishes at $x = 0$ and at $x = \frac{1}{2}$ (because $\sin\frac{\pi}{6} = \frac{1}{2}$), so these are the only nonnegative roots; by oddness $x = -\frac{1}{2}$ is the only negative one. Fixed points: $(0,0)$, $(\frac{1}{2},0)$, $(-\frac{1}{2},0)$.

$Df = \operatorname{diag}\big(\frac{\pi}{3}\cos\frac{\pi x}{3},\ \frac{1}{2}\big)$.

- $(0,0)$: eigenvalues $\pi/3 \approx 1.047 > 1$ and $\frac{1}{2}$: **saddle**.
- $(\pm\frac{1}{2},0)$: eigenvalues $\frac{\pi}{3}\cdot\frac{\sqrt{3}}{2} = \frac{\pi\sqrt{3}}{6} \approx 0.907$ and $\frac{1}{2}$: **sinks**.

Orbits. Always $y_n \to 0$. For x: after one step $x_1 \in [-1,1]$, and s is increasing on $[-1,1]$, so the orbit is monotone and converges to a fixed point. $s(x) > x$ on $(0,\frac{1}{2})$ and $s(x) < x$ on $(\frac{1}{2},1]$, so any $x_1 > 0$ goes to $\frac{1}{2}$; by symmetry any $x_1 < 0$ goes to $-\frac{1}{2}$. Therefore:

- if x_0 is a multiple of 3 (so $x_1 = 0$), the orbit goes to $(0,0)$;
- if $\sin(\pi x_0/3) > 0$, i.e. $x_0 \in (6k, 6k+3)$, it goes to $(\frac{1}{2},0)$;
- if $\sin(\pi x_0/3) < 0$, i.e. $x_0 \in (6k-3, 6k)$, it goes to $(-\frac{1}{2},0)$.

Exercise 2.7

Here $b = 0.3$, and $(1-b)^2 = 0.49$. For $\lambda^2 + B\lambda + C = 0$, both roots have modulus < 1 iff $|C| < 1$, $1 + B + C > 0$ and $1 - B + C > 0$ (Jury conditions).

(a) Fixed points satisfy $x^2 + 0.7x - a = 0$, which has real roots iff $a > -0.1225$. For a fixed point, $B = 2x$ and $C = -0.3$, so it is stable iff $|x| < 0.35$. The larger root $x_+ = \frac{1}{2}(-0.7 + \sqrt{0.49 + 4a})$ satisfies $x_+ > -0.35$ always, and $x_+ < 0.35$ iff $a < 0.3675$. The smaller root $x_- < -0.35$ always, and then one eigenvalue exceeds 1 in magnitude while the other has modulus < 0.3, so it is always a saddle. Hence one sink and one saddle occur exactly for

$$-0.1225 < a < 0.3675.$$

(b) A period-two orbit exists iff $a > \frac{3}{4}(0.7)^2 = 0.3675$. Its Jacobian product has $\det = b^2 = 0.09$ and trace $T = 4x_1x_2 + 2b = 4(0.49 - a) + 0.6$. Stability requires $1 - T + b^2 > 0$ and $1 + T + b^2 > 0$. The first gives $a > 0.3675$; the second gives $a < (1-b)^2 + \frac{1}{4}(1+b)^2 = 0.9125$. So there is a period-two sink exactly for

$$0.3675 < a < 0.9125.$$

Exercise 2.8

The image of the unit disk under A is an ellipse whose semi-axes are the singular values of A (square roots of the eigenvalues of A^TA), directed along the left singular vectors. Its area is $\pi|\det A|$.

(a) $A = \begin{pmatrix} 2 & 0.5 \\ 2 & -0.5 \end{pmatrix}$. $A^TA = \mathrm{diag}(8,\, 0.5)$, so the singular values are $2\sqrt{2}$ and $1/\sqrt{2}$, with $Ae_1 = (2, 2)$ and $Ae_2 = (0.5, -0.5)$. The ellipse has semi-axis $2\sqrt{2}$ along $(1, 1)$ and semi-axis $1/\sqrt{2}$ along $(1, -1)$:

$$\frac{(x+y)^2}{16} + (x-y)^2 = 1, \qquad \text{area} = \pi\,|\det A| = 2\pi.$$

(b) $A = \begin{pmatrix} 2 & 1 \\ -2 & 2 \end{pmatrix}$. $A^TA = \begin{pmatrix} 8 & -2 \\ -2 & 5 \end{pmatrix}$ has eigenvalues 9 and 4, so the singular values are 3 and 2. The ellipse has semi-axis 3 along $(1, -2)/\sqrt{5}$ and semi-axis 2 along $(2, 1)/\sqrt{5}$:

$$\frac{(x-2y)^2}{45} + \frac{(2x+y)^2}{20} = 1, \qquad \text{area} = \pi\,|\det A| = 6\pi.$$

Exercise 2.9

$\det A = 1$, so $A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$, which has integer entries. The inverse map is

$$S^{-1}(x, y) = (x - y,\ -x + 2y) \pmod 1.$$

Check: $A^{-1}A = \begin{pmatrix} 1\cdot 2 - 1\cdot 1 & 1\cdot 1 - 1\cdot 1 \\ -2 + 2 & -1 + 2 \end{pmatrix} = I$. Moreover, $S(v) = Av - k$ for some integer vector k, so $S^{-1}(S(v)) = A^{-1}(Av - k) = v - A^{-1}k \equiv v \pmod 1$, since $A^{-1}k$ is an integer vector. Similarly $S(S^{-1}(v)) \equiv v$.

Exercise 2.10

$S(v) = v$ means $(A - I)v \in \mathbb{Z}^2$.

(a) Take $A = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$, $x = \frac{1}{2}$, $y = \frac{\sqrt{2}}{2}$. Then $S(x, y) = (\frac{3}{2},\, y) \equiv (\frac{1}{2},\, y) = (x, y)$. The eigenvalues of A are 3 and 1.

(b) Take $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, $x = \sqrt{2} - 1$, $y = 2 - \sqrt{2}$, so that $x + y = 1$. Then $A(x, y) = (x + (x + y),\ y + (x + y)) = (x + 1,\, y + 1) \equiv (x, y)$. The eigenvalues of A are 3 and 1.

In both cases A has the eigenvalue 1, as Step 4 of Challenge 2 predicts.

Exercise 2.11

Fixed points satisfy $Mx + a = x$, i.e. $(I - M)x = a$. This has exactly one solution (namely $x = (I - M)^{-1}a$) for every a iff $I - M$ is invertible, i.e. iff 1 **is not an eigenvalue of** M. (If 1 is an eigenvalue, then for each a there are either no fixed points or infinitely many.)